

Multiple Choice

1. **Answer:** B

Options: A) names for the layers of gaseous and metallic hydrogen deep within the planet.; B) alternating bands of rising and falling air at different latitudes.; C) alternating regions of charged particles in Jupiter's magnetic field.; D) regions of the plasma torus created by ions from Io's volcanoes

Note: Correct option: B - alternating bands of rising and falling air at different latitudes.

2. **Answer:** C

Options: A) Brighter; B) Dimmer; C) Higher in the sky; D) Lower in the sky

Note: Correct option: C - Higher in the sky

3. **Answer:** C

Options: A) You will not be able to see these two stars at all.; B) You will see two distinct stars.; C) The two stars will look like a single point of light.; D) The two stars will appear to be touching looking rather like a small dumbbell.

Note: Correct option: C - The two stars will look like a single point of light.

4. **Answer:** D

Options: A) in the center of the asteroid belt; B) on orbits that cross Earth's orbit; C) surrounding Jupiter; D) along Jupiter's orbit 60° ahead of and behind Jupiter

Note: Correct option: D - along Jupiter's orbit 60° ahead of and behind Jupiter

4. **Answer:** A

Options: A) asteroids would orbit with a period half that of Jupiter; B) asteroids would orbit with a period twice that of Jupiter; C) asteroids would orbit with a period twice that of Mars; D) A and B

Note: Correct option: A - asteroids would orbit with a period half that of Jupiter

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Volcanic Heating'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'organisms living deep in the oceans around seafloor volcanic vents and in hot springs'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Moons contribute to all of the above.'

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'it switches polarity every 11 years'

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'There is no solar day Planet X is tidally locked'

Long Form Response

11. **Answer:** It would be the same no matter where you are.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** An idealized physical object that absorbs all electromagnetic radiation.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key